

End-to-End Workflow (How iScream Works)

1. Onboarding & POPIA Consent
2. Voice Calibration (3 screams) + Sensitivity
3. Always-on Listening (Android foreground service)
4. SOS Trigger + GPS Location
5. Guardian Alerts (SMS/WhatsApp/Push)
6. Evidence Capture (Phase 2) + Audit Log
7. Resolution + Reporting

MVP Highlights

- ? Locked-phone detection on Android (foreground service)
- ? 3-sample scream calibration
- ? SOS + GPS + guardian alerts
- ? POPIA-aligned consent & auditability

Impact Focus

Designed for GBV response in South Africa: rapid alerts, location visibility, and privacy-first handling of sensitive data.

Revenue Model (DTI-ready)

- ? Subscriptions (individual/family/community)
- ? Sponsored safety programs (NGO/corporate/municipal)
- ? Enterprise licensing + premium evidence storage

Technical Architecture (Build + Data Flow)

Mobile: React Native + Expo

Voice: On-device calibration + thresholds (MVP)

Backend: Firebase Auth + Firestore + Storage + FCM

Maps: GPS + reverse geocode

Notifications: FCM + SMS/WhatsApp (Phase 2)

Security & Compliance

? POPIA consent at onboarding

? Data minimisation & encrypted storage

? Audit logging on sensitive events

? Role-based access for staff portals

Phase Roadmap

Phase 1: Locked-phone Android MVP + SOS flow

Phase 2: Evidence recording + Support Centre

Phase 3: Watch integration + SAPS/EMS workflows

DTI Proof Points

? Safety impact (GBV response, faster alerting)

? Local jobs: operators, support, onboarding

? Local data residency & compliance readiness